Hands-on Exercise 1: SQL

1. Introduction

This document contains hands-on and practice exercises designed to help students understand and apply Structured Query Language (SQL) concepts using two relational databases: a **Company Database** and a **University Database**. The exercises cover database creation, schema implementation, data manipulation, and query formulation using SQL.

Structure of the Document

- **Part 1: Installation and Setup**
- **Part 2: Database Schema Implementation**
- **Part 3: Basic SQL Operations**
- **Part 4: Practice Queries**

Instructor: Dr. Yogesh K Meena

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2. Installation and Setup

Step 1: Download and Install MySQL Workbench

- https://dev.mysql.com/doc/workbench/en/wb-installing.html
- https://dev.mysql.com/downloads/workbench/

Step 2: Database Implementation
Implement the Company Database and the University Database as per the given schemas. The SQL files (`company_ddl.sql`, `company_dml.sql`, `university_ddl.sql`, `university_practice.sql`) are provided in the course plan.

Table 1: Employee

emp_id	first_name	last_name	birth_date	sex	salary	super_id	branch_id
100	David	Wallace	1967-11-17	M	250000	NULL	1
101	Jan	Levinson	1961-05-11	F	110000	100	1
102	Michael	Scott	1964-03-15	M	75000	100	2
103	Angela	Martin	1971-06-25	F	63000	102	2
104	Kelly	Kapoor	1980-02-05	F	55000	102	2
105	Stanley	Hudson	1958-02-19	M	69000	102	2
106	Josh	Porter	1969-09-05	M	78000	100	3
107	Andy	Bernard	1973-07-22	M	65000	106	3
108	Jim	Halpert	1978-10-01	M	71000	106	3

Table 2: Branch

branch_id	branch_name	mgr_id	mgr_start_date
1	Corporate	100	2006-02-09
2	Scranton	102	1992-04-06
3	Stamford	106	1998-02-13

Table 3: Client

client_id	client_name	branch_id
400	Dunmore Highschool	2
401	Lackawanna County	2
402	FedEx	3
403	John Daly Law, LLC	3
404	Scranton Whitepages	2
405	Times Newspaper	3
406	FedEx	2

Table 4: Works_With

emp_id	client_id	total_sales
105	400	55000
102	401	267000
108	402	22500
107	403	5000
108	403	12000
105	404	33000
107	405	26000
102	406	15000
105	406	130000

Table 5: Branch Supplier

branch_id	supplier_name	supply_type
2	Hammer Mill	Paper
2	Uni-ball	Writing Utensils
3	Patriot Paper	Paper
2	J.T. Forms & Labels	Custom Forms
3	Uni-ball	Writing Utensils
3	Hammer Mill	Paper

3. Some Basic SQL Queries

3.1. Creating a Database

```
CREATE DATABASE database_name;
```

3.2. Dropping a Database

```
DROP DATABASE database_name;
```

This command deletes an entire database, including all of its tables, data, and other objects. **Note:** This operation cannot be undone.

3.3. Creating a Table

```
CREATE TABLE table_name
(
    column1 datatype PRIMARY KEY,
    column2 datatype NOT NULL,
    column3 datatype UNIQUE,
    column4 datatype,
    column5 datatype CHECK (column5 > 0),
    column6 datatype,
    column7 datatype,
    CONSTRAINT fk_example
        FOREIGN KEY (column7)
        REFERENCES other_table(other_column)
);
```

3.4. Deleting a Table or Data

To delete an entire table:

```
DROP TABLE table_name;
```

To delete all records from a table (without deleting the table):

```
DELETE FROM table_name;
```

3.5. Copying a Table

```
CREATE TABLE new_table_name AS
SELECT * FROM source_table_name;
```

3.6. Adding a Column to a Table

```
ALTER TABLE table_name
ADD column_name datatype;
```

3.7. Modifying the Datatype of a Column

```
ALTER TABLE table_name
MODIFY COLUMN column_name new_datatype;
```

4. Select Queries

4.1. Select All Columns from a Table

```
SELECT * FROM table_name;
```

4.2. Select Specific Columns

```
SELECT column1, column2
FROM table_name;
```

4.3. Select with a Condition (WHERE Clause)

```
SELECT column1, column2
FROM table_name
WHERE condition;
```

4.4. Select with Multiple Conditions (AND / OR)

```
SELECT column1, column2
FROM table_name
WHERE condition1 AND condition2;
```

4.5. Select with Sorting (ORDER BY)

```
SELECT column1, column2
FROM table_name
ORDER BY column_name ASC;
```

4.6. Select with Limit

```
SELECT column1, column2
FROM table_name
LIMIT number;
```

4.7. Select Distinct Values

```
SELECT DISTINCT column1
FROM table_name;
```

5. Practice Queries (University Database)

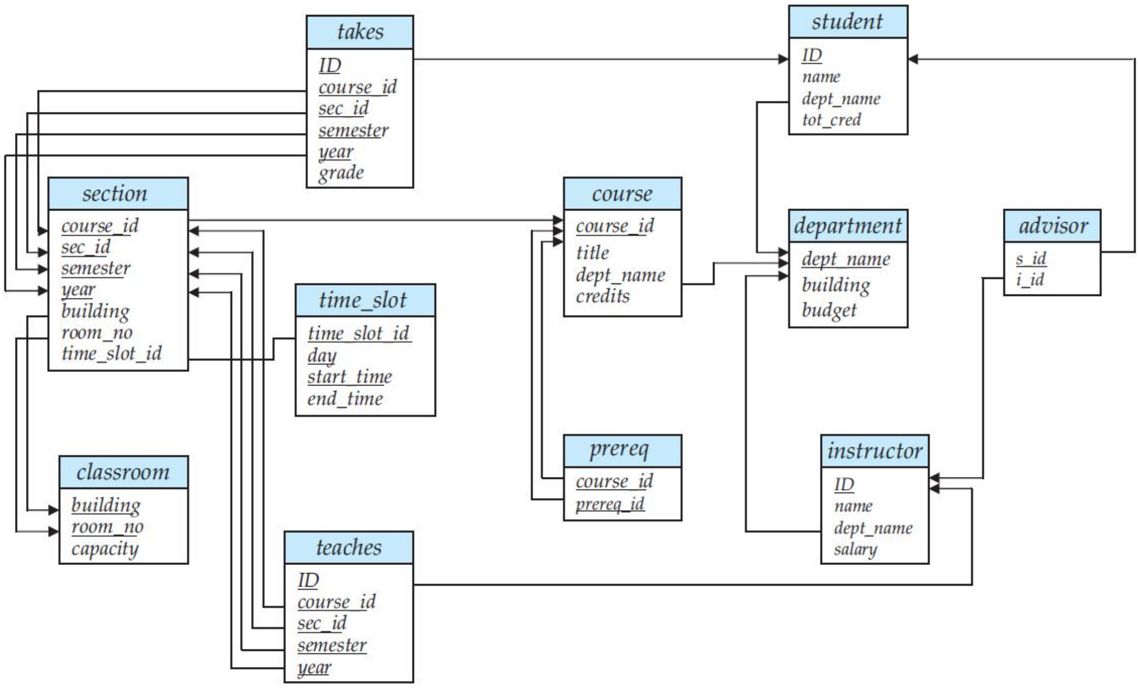

Figure 1: University database schema diagram

a. Find the titles of courses offered by departments other than Computer Science that have more than 3 credits.

b. Retrieve the IDs of instructors who have taught at least one student from the Computer Science department.

c. Find the second-highest salary earned by any instructor.

d. List the names and salaries of instructors whose salary is greater than the average salary of all instructors.

e. Find the total enrollment per course for courses offered in Fall 2017.

f. Identify the course IDs whose enrollment exceeded 50 students in Fall 2017.